

TRI PHONE: a pocket computer you can audit

$$\varphi^2 + 1/\varphi^2 = 3$$

Flagship course: a ternary FPGA pocket computer — and an SBC — running TRIOS with the TRI CLAW agent. Every layer auditable: from bitstream to the agent's word. Mesh connectivity, not cellular.

Rung v0 is in progress: TRI CLAW, a gateway concept. Tokens/s on the target part are NOT measured — said here, not discovered in due diligence.

PROBLEM

Low precision moved into silicon. Trust did not.

Precision became a hardware design variable.

Four bits went mainstream in one hardware generation; the conformance standard (IEEE P3109) is not ratified. Exactly one thing cannot be bought: proof that a decoder matches its spec bit for bit.

And at the same time: the agent moved onto the user's desk.

The agent-in-a-box already sells (a Jetson appliance with an agent stack, €549). Closed silicon, no attestation, no receipts: what the agent did cannot be verified even in principle.

The hole is at the intersection.

The verifiable device exists (Precursor). The agent exists (ClawBox). Ternary LLMs on FPGAs exist (in papers). Nobody ships a verifiable agent.

PRODUCT

A ladder: each rung sells alone and feeds the next

Rung	What it is	Hardware	Status
v0 · TRI CLAW	an agent you can audit: attestation at boot (Phi), a BLAKE3 receipt per action (Euler), a ternary engine with an empty DSP column (Gamma)	stock Artix-7 200T board; zero custom hardware	concept published; tok/s unmeasured
v1 · TRI-NET NODE	v0 + mesh radio with coverage seals: bytes across two radio hops byte-exact, the seal agreed at 3 points	Pluto/uSDR class; a partner hardware path	demo exists; throughput unmeasured
v2 · TRI PHONE — FLAGSHIP	pocket device: display, keyboard, battery, mesh; TRIOS + TRI CLAW; every layer auditable	a Precursor-class device of our own	a vision; open questions named

The three dies that never reached the foundry are the three layers.

TRI-1 Phi / Euler / Gamma: RTL green in CI, GDS built by the open flow, conformance vectors, Apache-2.0, DOI 10.5281/zenodo.19227877. No silicon exists — and none is needed: the layers ship as gateware.

An agent you can audit — the market’s empty intersection

Who	What	What is missing
Agent boxes (Jetson)	agent appliance, €549	no attestation, no receipts, no open bitstream
Precursor	a verifiable FPGA pocket device; >\$220k on Crowd Supply	no AI
TeLLMe / TerEffic / LUT-LLM	ternary LLM engines on edge FPGAs	papers, not products; verification not the point
TRI CLAW	the intersection of all three	a concept + three gateway layers; tok/s unmeasured

The open flow ends at the 7-series. Here that is a feature.

Precursor chose the 7-series as the most-studied silicon. An agent that sells trust targets silicon one can productively distrust. That flow carries 27 of our merged fixes.

Under the ladder, the rig: we sell proof, not a format

Implementation — synthesisable decoders and arithmetic cores.

RTL that passed the open flow on real hardware ships with the same synthesis commands that produced the numbers. No Vivado in the loop.

The conformance rig — bit-exact vectors plus an independent second oracle.

Copying a format is an evening's work; reproducing a conformance corpus a chip team will trust is not. That is the moat.

Theory that says which formats can exist at all.

T25 pins the three-symbol weight alphabet by uniqueness; T27 enumerates the multiplier-free scales. T26: in $\mathbb{Z}[\varphi]$ the network's linear part computes with zero rounding error — machine-checked in Coq. A lattice property, not a φ advantage.

TRI CLAW stands on this rig.

Attestation, receipts and the ternary engine are the same RTL blocks, the same vector discipline. The product does not replace the conformance business — it is its storefront.

MEASURED

What exists before any money

Every figure below came from the open toolchain on a named part. No frequencies are claimed: see the boundary line.

66 LUT

MEASURED

GFTernary decoder, XC7A200T; bare wire is 112 LUT

38×

MEASURED

adder: 397 LUT vs 15,251 for tekum8

1.4–1.9× / 3.2–4.1×

MEASURED

unfolded ternary node vs int4/int8, fan-in 8–64, post-synthesis; series non-monotonic: 56.2→75.4→66.8→68.1

0

MACHINE-CHECKED IN COQ

rounding error on the linear path in $Z[\varphi]$, T26

52 / 83

PROVEN

theorems in the paper / formats in the catalogue

27

MERGED UPSTREAM

fixes merged into nextpnr-xilinx — the flow that builds all of this

2

RADIO HOPS

byte-exact; the coverage seal agreed at 3 points: 0x9DBE2510

9

THEIRS, NOT OURS

TeLLMe tok/s on a KV260 at 7 W — existence, not transfer

The claim boundary, drawn by us

No silicon. No ternary fabric was used. Energy unmeasured. TRI CLAW tok/s on the A200T unmeasured. TRIOS has never run on an embedded core. Node figures are post-synthesis on an unfolded circuit, not post-P&R on a folded one.

MARKET

Bottom-up, from other people's tills

A Precursor-class device crowdfunds: >\$220k, <500 units, \$512–768.

Verifiability alone carried the campaign. Crowd Supply is full-service (~12% fee, Matching Funds 50–100% on top, Mouser logistics).

The audience is already counted by neighbours: 190 uSDR backers $\times$ \$339–899 = \$128k.

Open-radio and open-hardware engineers — the same people who buy Precursor and agent boxes.

The IP layer underneath does not go away.

The semiconductor IP market is \$9.8B (2025); a licence proxy is ~\$1.2M; royalties at rates Arm discloses. The product is the rig's storefront, not its replacement.

Five revenue lanes, all sold to teams, not end users

Layer	What the customer gets	Price / cycle
TRI CLAW product (v0)	gateway+stack on a stock board; a Crowd Supply campaign after tok/s is measured	\$500–800/unit, Precursor precedent
Core licence	a synthesisable decoder or core: RTL + synthesis commands	per project; proxy ~\$1.2M / 6–12 mo
Conformance rig	bit-exact vectors, second oracle, seeds and flow	annual subscription
Royalties	per unit of shipped silicon	Arm discloses ~5% as comparable
Measurement as a service	customer sends RTL — measured on the same rig and seeds	fixed / 30–60 days

The moat is the rig and the discipline, not patents.

Preprints and open RTL create publication priority, not exclusivity. The defensible asset is a corpus a chip team will trust — and the public record of withdrawn claims.

TEAM

One engineer, one rig, a public record of mistakes

Dmitrii Vasilev — founder.

FPGA/RTL and hardware-AI engineer. Runs the whole line: spec → RTL → open flow → measurement → paper. Two public preprints (arXiv 2606.05017, 2606.09686), an 83-format catalogue, a third paper with 52 theorems (Coq where stated), 27 merged nextpnr-xilinx fixes, a rig of three ALINX AX7203 boards.

The rare part is neither the maths nor the RTL.

One person carries a claim from theorem to placed-and-routed number and publishes his own retractions. The same discipline is baked into the product: receipts, attestation, an open bitstream.

Three steps, each externally checkable

1 · Measure tok/s of a small ternary model (0.1–0.7B) on the A200T.

Table-lookup matmul — the approach academia converged on. The number gets published whatever it turns out to be. This gates the whole ladder.

2 · A minimal TRIOS on a weak core + a 90-second demo.

Boot → attestation → one agent task → one receipt → the receipt verified on a different computer.

3 · Apply to Crowd Supply with the number in hand.

The shelf neighbour is Precursor. The open questions (the “claw” name space is crowded; entity and platform compliance; battery unmeasured) go into the campaign as its first section, as is our habit.

Three things worth remembering.

Nobody ships a verifiable agent; our three unfabricated dies are that agent’s ready-made layers; and everything we do not know is written on this slide, not hidden.